Maya v2: how much power to buy the day before

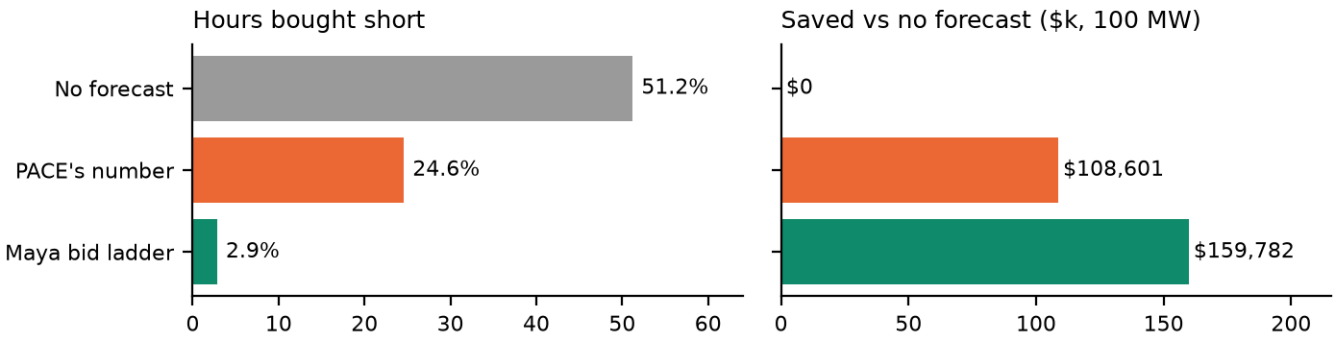
Maui Hiva Innovations | Purdue MS Data Science capstone, Project T | updated October 21, 2026 | live dashboard: maya-v2-buy-ahead.streamlit.app

What it does

Every morning before the 10 am Mountain day-ahead deadline, Maya forecasts tomorrow's net load for the PacifiCorp East area (demand minus Milford, Utah renewable output) as a range, not a single number, and turns it into a buy plan. Buying short means paying the real-time price for the gap, so the plan is built to be short rarely and on purpose. It also flags hours at risk of a price spike, suggests when to run flexible load, and gives a 7-day outlook of daily net load for weekly planning.

Results

	Maya	Comparison
Best-guess error, 2026 (MWh per hour)	**201**	PACE's own forecast: 343 (41% higher)
Live, 48 unseen days: hours bought short	**3.9%**	PACE's number: 30%
Live best-guess error (MWh per hour)	**300**	PACE: 480
Saved vs buying with no forecast*	**$159,783**	90% range $64,675 to $265,417
Saved vs buying PACE's number*	**$51,181**	90% range $13,556 to $90,160
Load shifting, 5 MW moved 4 h a day*	**$88,862**	92% of the perfect-hindsight value; saved money on 100% of days



*Real CAISO day-ahead and real-time prices for the PACE area, May 01 to Oct 20, 2026 (172 days), at a placeholder size of 100 MW until Maui Hiva confirms its load. 90% ranges from a bootstrap over whole days.

How it works

1. A calibrated range. Two models (gradient boosting and an LSTM) forecast levels from P50 (best guess) to P98, using ECMWF weather forecasts (Milford plus eight load-center cities), holidays and school breaks, and only data available at the deadline. Each level is re-checked daily, so P95 really is short about 5% of the time.

2. Risk and the bid ladder. A daily Green / Yellow / Red tier sets the always-bought amount (P95, or P98 on risky days). An hourly spike-risk model sets a price limit for an extra block up to P98, bought only when day-ahead is cheaper than expected real-time.

3. Load shifting. Expected hourly prices (last 7 days of day-ahead results) show which hours to move flexible load out of and into.

4. Runs itself. A daily job pulls EIA demand, ECMWF weather and CAISO prices, forecasts, and scores the track record. A monthly check retrains the models and switches only if the new one is better on unseen days.

Limits

Day-ahead prices for PACE exist only since May 2026 (CAISO EDAM), so dollar results rest on a short window. Milford renewable output is modeled, not metered. Maui Hiva's own load is assumed to follow PACE's pattern. On ordinary days the buy strategies differ by cents per MWh; Maya's main value is being short far less often, which is what matters on spike days.

Dashboard: maya-v2-buy-ahead.streamlit.app | Code and data: github.com/RendellRich/maya-forecasting-v2